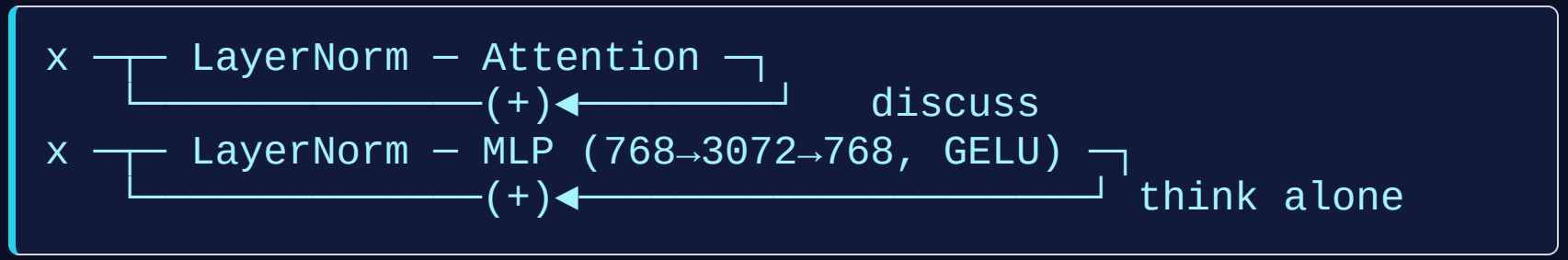
Assemble the GPT

Module 6 · Zero to LLM

A block = discuss, then think



Residual = notes in the margin · Pre-norm = calm breath first

LayerNorm check

`[102.5, 98.3, 105.7, 95.2]` → `≈[0.52, −0.53, 1.32, −1.31]` — **sums to 0**

GELU (tanh approx) in GPT-2 · SwiGLU in Llama-style models

The whole model

token emb + pos emb → 12 × Block → LayerNorm → lm_head (tied to token emb)

124,439,808 parameters (= Hugging Face GPT-2)

Verify against a reference

- HF stores 4 linear weights transposed (Conv1D)
- max logit diff ≈ 1e-4 → it *is* GPT-2
- `bias=False` on attention → biases silently skipped → "foxeseseses…"

Same code, bigger numbers

model	layers	d_model	heads	params
small	12	768	12	124 M
medium	24	1024	16	355 M
large	36	1280	20	774 M
XL	48	1600	25	1.5 B

Next: M07 — train our own